18 A uniform copper rod of cross-sectional area 8.0×10^{-6} m^2 has 8.5×10^{28} conduction electrons per cubic metre. A current flows through the rod when a potential difference is applied across it.

Given that the drift velocity of electrons in the rod is 2.3×10^{-5} m s^{-1}, what is the current in the rod?

A 0.25 A

B 0.40 A

C 2.5 A

D 4.0 A